

Accurate and Dynamic Network Topology Representation based on Functional Block-based Disaggregation for a Common Reference of Network Digital Twins

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Abstract We develop a data platform integrating component-level topology and port-based reservations using a relational database for optical network digital twins. Field-testbed experiments evaluate the execution times of digital- and physical-space update operations, demonstrating timely and consistent synchronization and its role as a real-time common reference. ©2026 The Author(s)

Introduction

Agile, flexible, and intelligent optical network management and control are increasingly indispensable to accommodate the rapidly growing volume, heterogeneity, and dynamic behaviour of communication traffic driven not only by human-centric services but also by AI-centric workloads. In this context, optical network digital twin technologies have been actively investigated, specifically for physical layer modelling, real-time monitoring, and automatic control and orchestration, including AI/ML assisted operational flows [1-8]. However, existing approaches are often implemented separately, using different datasets, assumptions, and data formats, which makes it difficult to integrate them into end-to-end digital twin workflows. A unified data management platform that serves as a common reference remains a critical missing link. Such a platform should consistently integrate all relevant data and dynamically reflect changes in the physical network, supporting much more frequent operations driven by AI agents than human-operated workflows.

To provide such a common reference, we develop an optical network topology and resource management system based on an RDBMS (relational database management system). The system adopts a functional block-based disaggregation (FBD) model that explicitly models individual optical components, specifies intra-node structures, and embeds component-wise physical parameters [9,10], enabling a topology/resource representation sufficiently accurate for digital twin operation. Consistent data management under concurrent accesses and frequent updates is achieved by relying on ACID (atomicity, consistency, isolation, and durability) properties of the RDBMS. We implement the proposed system on a field-deployed optical fibre testbed and evaluate operation times across digital and physical domains, covering path search, topology update, and path establishment/removal operations. The results demonstrate that the proposed system enables practically real-time data access and achieves timely and consistent synchronization between the digital- and physical-spaces.

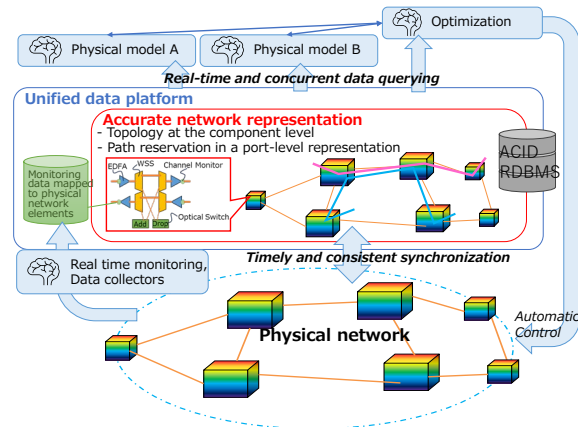


Fig. 1: Overview of a unified data platform.

System Overview and Implementation

Fig. 1 illustrates an overview of the proposed unified data platform, which serves as a common reference shared across digital twin modules. By using the FBD model, the platform maintains a network representation that accurately reflects the physical optical network topology at the component level and manages path reservation states using a port-level representation. Based on this accurate topology representation, other relevant data—such as time-series monitoring data—can be consistently aggregated and explicitly mapped to individual physical components. While many existing studies on digital twin focus on individual devices or on optical paths formed by a sequence of device ports, the proposed platform provides comprehensive and consistent information associated with both individual components and end-to-end optical paths. As a result, heterogeneous digital twin modules can be integrated on top of this unified data platform.

Fig. 2 shows an overview of the system implementation of the proposed topology management system. The system is designed to maintain an accurate network representation and to support timely and consistent synchronization between the digital- and physical-spaces through database integration. For intuitive drawing of optical network topologies, KiCad, an open-source electronic design automation software, is employed

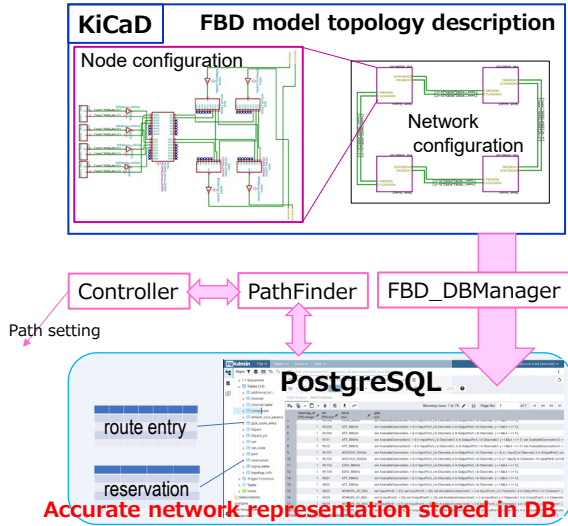


Fig. 2: System implementation overview.

as a graphical user interface (GUI). Optical component libraries have been developed, and the necessary information, including physical parameters, are embedded in individual component instances [10]. The FBD_DBManager module bridges the topology data created in KiCad to a PostgreSQL database. When a topology update occurs, FBD_DBManager detects the differences between the current and updated KiCad representations and incrementally updates the database entries with appropriate version identifiers.

The PathFinder module computes optical path routes using an integer linear programming (ILP) optimization solver and maintains a persistent connection to the topology database to ensure consistency with up-to-date network states. For efficient path computation processes, PathFinder generates intermediate data files prior to receiving optical path computation requests. Upon topology updates, PostgreSQL notifies PathFinder of the changes, triggering automatic regeneration of the intermediate data files. As a result, no manual reset operations are required. The Controller module configures the network equipment to establish the computed optical paths in the physical network. If path configuration fails in the physical network, PathFinder deletes the corresponding path reservation from the database.

The underlying topology description scheme of the FBD model, including its path computation, interoperability with standardized network models, and GNPY integration, is described in [9,10] and references therein. In addition, PostgreSQL is supported by existing implementations of the Model Context Protocol (MCP) [11], making the proposed system well-suited for integration with LLM-based AI agents.

Although the complete database schema is not fully described here due to space limitations, the most representative tables are outlined below.

For component modeling, a *component table* manages individual component instances, while a *libparts table* stores default component library information. This KiCad data-format-aware design enables consistent and reliable handling of online topology updates via KiCad GUI.

For PathFinder-related data, a *reservation table* maintains basic reservation information, such as reservation identifiers, source ports, and destination ports, while a *glpk_route_entry table* stores fine-grained path descriptions. These descriptions include an ordered list of port-by-port route entries and the wavelength channel assigned to each section. This port-level path representation allows path computation results to be directly translated into device-level configuration commands. As a result, the proposed system eliminates the need for an additional hardware abstraction layer (HAL) required in conventional disaggregated networks.

To ensure consistent and safe topology updates, foreign key constraints are designed across topology and reservation-related tables. To be precise, attempts to remove or modify network components that are currently associated with active path reservations result in database-level constraint violations. This mechanism prevents inconsistent topology states and provides a robust foundation for safe operational procedures, such as path-aware component isolation, failure impact analysis, and recovery planning. This capability becomes increasingly important as network scale and operational frequency grow.

Experimental Demonstration

Fig. 3 illustrates the experimental testbed, which consists of four sites: SiteA, SiteB, SiteC, and SiteR. SiteA–SiteR, SiteB–SiteR, and SiteC–SiteR are interconnected via 30-km and 40-km spooled fibers and a 27-km field-deployed fiber, respectively. Each of SiteA, SiteB, and SiteC is equipped with a disaggregated ROADM system comprising two WSSs and two EDFAs, along with two white-box packet transponders. SiteR consists of three disaggregated ROADMs forming a non-blocking 3×3 wavelength cross-connect.

The detailed network topology information is managed by the proposed database system, which is implemented on an off-the-shelf laptop computer. Through the PathFinder and Controller modules, a total of 18 optical paths were provisioned, as summarized in Tab. 1, and their corresponding optical spectra are shown in Fig. 3.

To evaluate the real-time data access capability of the proposed system, we measured the execution time of database queries for path identification operations. Queries were issued to identify optical paths traversing specific ports or components under assumed network failure

scenarios. Six failure scenarios were considered: (1) node egress port failure, (2) intra-node connection port failure, (3) add/drop port failure, (4) WSS component failure, (5) EDFA component failure, and (6) white-box packet transponder failure. The failure locations are indicated in Fig. 3, and the affected optical paths for each scenario are summarized in Tab.1. Tab. 2 summarizes the average, minimum, and maximum execution times, with each query repeated ten times. For all evaluated scenarios, query execution times of less than one second were achieved. While the absolute execution time depends on the database size, the underlying database schema and query structure are independent of the network scale; the impact of network size on query performance will be investigated in future work.

We then measured the database execution time for a topology update operation. Assuming a new transponder installation, we updated the topology information with KiCad and observed that the database update process was completed within 15 seconds, including 4 seconds for KiCad data export, 5 seconds for database updates, and 5 seconds for intermediate data file regeneration. Given that physical-space operations such as equipment replacement or installation typically require several hours, the observed digital-space update execution time is sufficiently short for practical network operations.

To evaluate the end-to-end operation times, we performed optical path deletion operations according to failure scenario (1), followed by re-

establishment of the corresponding paths. Fig. 4 shows an excerpt from the operation log. Deleting 13 optical paths required 51 seconds, while re-establishing the same 13 paths took 6 minutes. The dominant contributors were network equipment control via the NetConf API, the ILP-based PathFinder computation, and optical power adjustment. Owing to the limited refresh rate of the channel power monitor in the employed ROADMs, a three-second waiting period was introduced before reading the power levels and adjusting the WSS attenuation to satisfy the receiver power range. As optical path provisioning involves sequential control actions, physical network operations typically span seconds to minutes. The necessity of reflecting such short-lived transitional states in the digital-space depends on future requirements for network-wide optimization and the adoption of AI-assisted control, leaving room for further discussion.

Conclusion

The experimental results demonstrate that the proposed system enables practically real-time data access and achieves timely and consistent synchronization between the digital- and physical-spaces, i.e., sufficiently faster update in the digital-space than physical topology changes. While several challenges remain, including integration with other digital twins and handling physical-space transactional operations, the proposed data platform will serve as a foundational building block of a digital twin data platform.

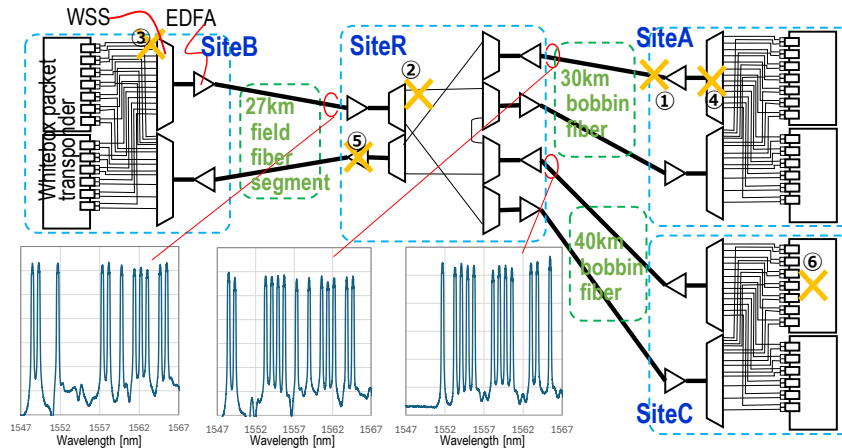


Fig. 3: Experimental setup including 27-km field fiber segment.

Tab. 2: Path Search Query Results

Failure scenario	# of affected paths	Query execution time [ms]		
		Min	Max	Average
①	13	228	421	320
②	6	157	679	315
③	1	189	471	320
④	13	227	700	481
⑤	11	211	539	359
⑥	6	192	542	282

Tab. 1: Provisioned Optical Path

id	Src/Dst sites	Ch #	Failure scenario					
			①	②	③	④	⑤	⑥
1	SiteA SiteB	WDM96_6	x	x	x	x		
2	SiteA SiteB	WDM96_12	x	x	x	x		
3	SiteA SiteB	WDM96_14	x	x	x	x		
4	SiteA SiteB	WDM96_24	x	x	x	x		
5	SiteA SiteB	WDM96_44	x	x	x	x		
6	SiteA SiteB	WDM96_46	x	x	x	x		
7	SiteA SiteC	WDM96_8	x			x		
8	SiteA SiteC	WDM96_16	x			x		x
9	SiteA SiteC	WDM96_20	x			x		
10	SiteA SiteC	WDM96_28	x			x		x
11	SiteA SiteC	WDM96_30	x			x		x
12	SiteA SiteC	WDM96_32	x			x		x
13	SiteA SiteC	WDM96_34	x			x		x
14	SiteB SiteC	WDM96_4		x		x		
15	SiteB SiteC	WDM96_10				x		
16	SiteB SiteC	WDM96_18				x		
17	SiteB SiteC	WDM96_22				x	x	
18	SiteB SiteC	WDM96_38				x		

```
[2026-02-19 13:43:27] + curl -X DELETE http://localhost:33300/paths/urn:uuid:0433a7b5-b51e-4c38-bbd3-0c6c1fbb8918" , "status": "success"}
[2026-02-19 13:43:32] {"path_id": "urn:uuid:0433a7b5-b51e-4c38-bbd3-0c6c1fbb8918", "status": "success"}
~~~~~snipped during path deletion operations~~~~~
[2026-02-19 13:44:15] + curl -X DELETE http://localhost:33300/paths/urn:uuid:d42676f0-ff5a-4a95-b53a-9c8b4e37030c" , "status": "success"}
[2026-02-19 13:44:18] {"path_id": "urn:uuid:d42676f0-ff5a-4a95-b53a-9c8b4e37030c", "status": "success"}
~~~~~starting path re-establishment operations~~~~~
[2026-02-19 13:45:11] curl -sS -X POST http://localhost:33300/paths -H "Content-Type: application/json" -d '{"path_id": "urn:uuid:a6552ab8-fca6-4892-ba45-d0b29be6f2cc", "status": "success"}'
[2026-02-19 13:45:31] {"path_id": "urn:uuid:a6552ab8-fca6-4892-ba45-d0b29be6f2cc", "status": "success"}
~~~~~snipped during path re-establishment operations~~~~~
[2026-02-19 13:50:43] curl -sS -X POST http://localhost:33300/paths -H "Content-Type: application/json" -d '{"path_id": "urn:uuid:0039a972-0041-4aa3-9c16-663c917fb364", "status": "success"}'
[2026-02-19 13:51:16] {"path_id": "urn:uuid:0039a972-0041-4aa3-9c16-663c917fb364", "status": "success"}
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Fig. 4: Log excerpt of path removal/establishment operations.

Acknowledgements

This presentation is partly based on results obtained from projects, JPNP16007, commissioned by the New Energy and Industrial Technology Development Organization (NEDO) and CRONOS, Japan Grant Number JPMJCS25N3, supported by JST.

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